

Streams, Friends & the Many Ways to Build an Object

Quiz 03 — `operator<<`, `friend`, member initializer lists

Name _____

Date _____

array[0] Find the Bug

identify & fix

COMPILES, PRINTS GARBAGE

```
class Point {
    int x, y;
public:
    Point(int x, int y) {
        x = x;
        y = y;
    }
    friend ostream& operator<<(ostream& os, const Point& p);
};
ostream& operator<<(ostream& os, const Point& p) {
    return os << "(" << p.x << ", " << p.y << ")";
}
int main() {
    Point p(3, 4);
    cout << p;    // (0, 0) -- wrong!
}
```

Which line is the bug on, and why does it leave `x` and `y` unset? Then rewrite **just the constructor** so it works — use a member initializer list.

BUG & WHY

CORRECTED CONSTRUCTOR

array[1] Write the Code

friend + `operator<<`

STARTING POINT

```
class Fraction {
    int num, den;
public:
    Fraction(int n, int d) : num(n), den(d) {}
    // (a) add one line here
};
// (b) write the function here so cout << f prints "num/den"
```

(a) Add the **one line** inside the class that lets a non-member `operator<<` reach the private `num` and `den`. (b) Write the free function so that `cout << f;` prints `num/den` (e.g. `3/4`).

(A) + (B)